BASIC TERMINOLOGIES

1) PLAIN TEXT:- original message which has to be encrypted.

2)CiPHER TEXT:- encrypted message.

3)ENCRYPTION:- process of converting plain text into cipher text.

4)ENCRYPTION ALGORITHM:- plain text and secret key.

5)DECRYPTION:- recovering plain text from cipher text.

6)DECRYPTION ALGORITHM:- cipher text and secret key.

7)SECRET KEY:- it is used as an input in mathematical functions to encrypt plain text.

8)CRYPTOGRAPHY:- it is the method used to protect information and communication through the use of code, so that only those for whom information is intended can read and process it. The prefix “crypt” means “hidden” and the suffix “graphy” stands for “writing”.

9)CRYPTOANALYSIS:- it is the process of studying cryptographic systems to look for weaknesses and leak of information.

10)CRYPTOLOGY:- cryptology+cryptoanalysis.

TYPES OF ENCRYPTIONS:-

1)STREAM ALGORITHM: Data is encrypted as it streams instead of being retained in the system’s memory.

Bit by bit.

Valid only for short length messages.

2)BLOCK ALGORITHM: Set length of bits are encrypted in blocks of electronic data with the use of a specific secret key. As the data is being encrypted, the system stores data in its memory as it waits for a complete block.

THE OSI SECURITY ARCHITECTURE

The OSI security architecture is useful to manage task of providing security.

ITU-T4 recommended X.800 security architecture for OSI, defined as a systematic approach.

Focuses on security attacks, mechanism and service.

Security attack:- any action which can harm the security of information owned by an organization.

Security mechanism:- A process designed to detect, prevent and recover data from security attacks.
Security services:- processing and communication services that enhance the security of data processing systems.

KEY PRINCIPLES OF SECURITY/SECURITY SERVICES

1. Confidentiality-
Confidentiality ensures that no one can read messages except the intended receiver.
2. Authentication-Process to recognizing user's identity.
3. Integrity-no modification should be done during transmission.
4. Non repudiation- non repudiation prevents either sender or receiver from denying a transmitted message.
Sender and receiver can prove a message has been sent or received.
Prevents from DOS.
5. Access control-
The principle of access control determines who should be able to access what.
Authentication & Authorization
i) Authentication: Who is doing actual work
Eg:- ATM card
ii) Authorization: Access permission
What the user wants to do.
6. Availability- Refers to availability of information to authorized users.

SECURITY ATTACK:-

Security attacks are unauthorized actions against private , corporate or government IT assets in order to destroy them, modify them or steal sensitive data.
There are two types of security attacks:-
1) Active attack
2) Passive attack:

- The main goal of passive attack is to obtain unauthorized access to the information.

- Passive attacks are the attacks where the attacker indulged in unauthorized eavesdropping. The eavesdropping does not make any changes to the data or the system.
- Just monitoring the transmission or gathering information.
- No data modification.
- Unauthorized deletion of data.
- With passive attack, the attacker tries to collect or learn information from applications but does not affect the application itself.
- Denial of access to information for legitimate users.

KEY DIFFERENCE BETWEEN ACTIVE ATTACK AND PASSIVE ATTACK

- The active attack includes the modification of the message. While in passive attack, the attacker does not make any changes to the information.
- The active attack causes a huge amount of harm to the system. While passive attack does not cause any harm to the system resources.
- The active attack is a threat to the availability and integrity of the data. While passive attack is a threat to the data confidentiality.
- The attacked entity is aware of the attack in case of an active attack. On the other hand, in passive attack, the victim is unaware of the attack .
- The active attack is accomplished by gaining the physical control over the communication link to capture and insert transmission. On the other hand, in passive attack, the attacker just need to observe the transmission.